

◆ Zoho String + Array Mixed Problems (Level-2)

1. Anagram Grouping

- **Problem:** Given an array of strings, group all anagrams together.

Input:

```
arr = {"eat", "tea", "tan", "ate", "nat", "bat"}
```

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Output:

```
[["eat", "tea", "ate"], ["tan", "nat"], ["bat"]]
```

- **Constraints:** Case-sensitive, no built-in hashmaps.
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2. Longest Common Prefix

- **Problem:** Find the longest common prefix among an array of strings.

Input:

```
arr = {"flower", "flow", "flight"}
```

Output:

`"f1"`

- **Constraints:** $O(N \cdot M)$, avoid built-in substring methods where possible.
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3. Maximum Character Frequency Word

- **Problem:** Given an array of strings, find the word with the maximum frequency of a single character.

Input:

```
arr = {"apple", "banana", "grape"}
```

Output:

`"banana" (since 'a' occurs 3 times)`

4. Rearrange Words by Length

- **Problem:** Given an array of words, sort them by length while keeping relative order of same-length words.

Input:

```
arr = {"cat", "a", "apple", "dog", "on"}
```

Output:

```
{"a", "on", "cat", "dog", "apple"}
```

5. Word with Maximum Vowels

- **Problem:** From an array of strings, return the word with the maximum number of vowels.

Input:

```
arr = {"strength", "education", "quiz"}
```

Output:

```
"education"
```

6. Palindrome Pairs

- **Problem:** Find all index pairs (i, j) such that `arr[i] + arr[j]` is a palindrome.

Input:

```
arr = {"bat", "tab", "cat"}
```

Output:

```
[(0,1), (1,0)]
```

7. Remove Duplicate Characters in Words

- **Problem:** For each word in an array, remove duplicate characters and print.

Input:

```
arr = {"banana", "apple"}
```

Output:

```
{"ban", "apple"}
```

8. Find Word with Highest Sum of ASCII Values

- **Problem:** Given an array of strings, return the word with the maximum sum of ASCII values of its characters.

Input:

```
arr = {"abc", "zzz", "pq"}
```

Output:

```
"zzz"
```

9. Print All Substrings of Words

- **Problem:** For each word in an array, print all possible substrings.

Input:

```
arr = {"dog"}
```

Output:

```
d, o, g, do, og, dog
```

10. String Compression (Array of Strings)

- **Problem:** Compress each word in the array using counts.

Input:

```
arr = {"aabb", "aaabbc"}
```

Output:

```
{"a2b2", "a3b2c1"}
```

11. Print Words That Are Palindromes

- **Problem:** From an array of strings, print only palindromic words.

Input:

```
arr = {"madam", "racecar", "hello"}\
```

Output:

```
madam racecar
```

12. Concatenate Words Without Repetition

- **Problem:** Concatenate all words ensuring no duplicate letters across them.

Input:

```
arr = {"abc", "bcd", "cde"}
```

Output:

```
"abcde"
```

13. Rotate Array of Strings

- **Problem:** Rotate an array of words by **k** places.

Input:

```
arr = {"apple", "banana", "cherry", "date"}, k=2
```

Output:

```
{"cherry", "date", "apple", "banana"}
```

14. Find Word with Maximum Unique Characters

- **Problem:** From an array of strings, return the word with maximum unique letters.

Input:

```
arr = {"hello", "world", "zoho"}
```

Output:

```
"world"
```

15. Lexicographic Smallest Rotation

- **Problem:** Given a string, find its lexicographically smallest rotation (cyclic shift).

Input:

"bca"

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Output:

"abc"
